RAJAN SINGH

Data Analyst | Software Developer | AI/ML Engineer

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PROFESSIONAL SUMMARY

B.Tech Artificial Intelligence & Machine Learning student at DIT University with hands-on experience building AI/ML projects using Python and implementing machine learning algorithms from scratch. Strong foundation in Python, data science, machine learning, SQL and problem solving, with an interest in developing practical AI-powered applications.

EDUCATION

DIT University, Dehradun — B.Tech, Artificial Intelligence & Machine Learning

Expected Graduation: 2026

TECHNICAL SKILLS

Programming: Python, Java, C++
AI/ML: Artificial Intelligence, Machine Learning, Deep Learning, Neural Networks, PyTorch, Linear Regression, Supervised Learning, Genetic Algorithms
Data & Libraries: NumPy, Pandas, Scikit-learn, SQL, Jupyter Notebook, Matplotlib
Core: Data Preprocessing, Model Training, Regression, Algorithm Design, Tkinter GUI Development, Problem Solving, Git/GitHub

PROJECTS

AI Journal App

- Developed an AI-powered journaling application designed to help users create and interact with journal entries.
- Integrated AI functionality to make journal entries more interactive and useful, with focus on a simple user experience.
- Applied concepts of AI integration, application development and data handling.

Linear Regression Model From Scratch

- GitHub: github.com/jonathan6378/linear-regression-from-scratch
- Implemented a Linear Regression model from scratch without relying on a pre-built machine learning model.
- Applied the cost function and gradient-based optimization to train the model.
- Used Python and numerical computing techniques to understand and implement the internal working of a regression algorithm.

Smart Timetable Generator (Genetic Algorithm)

- GitHub: github.com/jonathan6378/smart-timetable-generator
- Designed and built a genetic algorithm-based timetable and job scheduling engine in Python as a final-year academic project.
- Built a Tkinter desktop GUI with SQLite integration for data persistence and Excel/PDF export of generated schedules.
- Used Matplotlib to visualize fitness scores and optimization progress across generations.

Mini GPT-Style Language Model

- Built a small GPT-style language model from scratch in PyTorch to learn transformer architecture internals.
- Implemented core components including tokenization, self-attention, and the training loop without relying on pre-built model libraries.
- Strengthened understanding of deep learning fundamentals and neural network training dynamics.

CERTIFICATIONS

- Python for Data Science, AI & Development — IBM, Coursera | 2024
- Introduction to Relational Databases (RDBMS) — IBM, Coursera | 2024
- Introduction to Contemporary Operating Systems and Hardware 1b — Illinois Tech, Coursera | 2024
- Goldman Sachs Job Simulation — Forage
- Artificial Intelligence Certificate

ADDITIONAL STRENGTHS

Problem Solving • Analytical Thinking • Machine Learning Fundamentals • Continuous Learning • AI Application Development